



Smart Waste Sorting Classifier

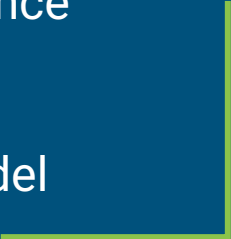


ITAI 1378 - Computer Vision Artificial Intelligence

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Project Tier: Tier 2 - Deep Learning Vision Model



The Problem

Waste sorting is often done manually, which can be slow and inaccurate.

Many people also throw recyclable items into regular trash because they cannot easily identify the correct category.

Who cares

- Recycling centers
- Waste management companies
- Environmental organizations
- Cities and municipalities

Why it is important

Incorrect waste sorting increases landfill waste and reduces recycling efficiency. An automated system could help classify waste correctly and support recycling efforts.

My Solution (Overview)

Solution

Develop a computer vision system that automatically identifies the type of waste from an image.

Workflow

Image of Waste → AI Model → Waste Category → Display Result

One-sentence description

A deep learning model will analyze waste images and classify them into recycling categories.

Technical Approach



Computer Vision Technique

Image Classification



Model

MobileNetV2 (Transfer Learning)



Framework

TensorFlow / Keras



Why this approach

MobileNetV2 is efficient, lightweight, and performs well on small to medium-sized datasets, making it ideal for this project.

Data Plan

Source

Garbage Classification Dataset from Kaggle.

<https://www.kaggle.com/asdasdasdas/garbage-classification?resource=download>

Dataset Size

~2,500 images.

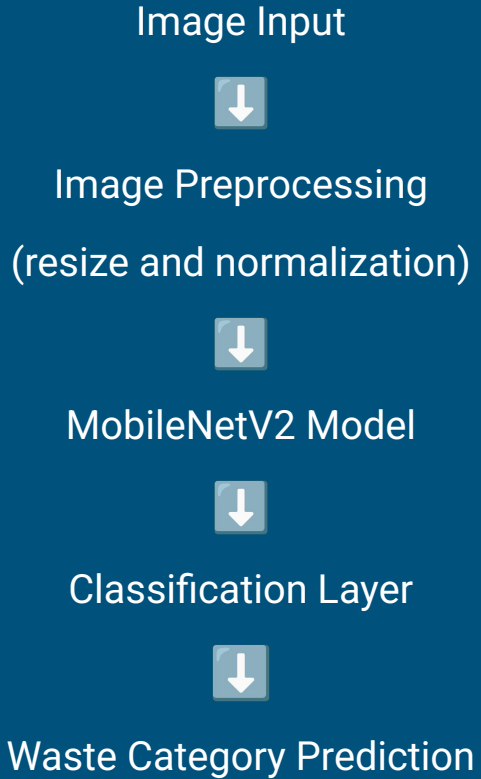
Classes

Plastic, Paper, Metal, Glass, cardboard, Trash

Data Preparation

1. Resize images to 224 × 224 pixels
2. Normalize pixel values
3. Split dataset into training and testing sets (80/20)
4. Apply data augmentation (rotation, flip, brightness)

System Diagram



Success Matrix

Metric Type	Measurement	Target
Primary Metric	Classification Accuracy	$\geq 90\%$
Secondary Metric	Prediction Speed	< 1 second
Additional Metric	Precision / Recall	$\geq 85\%$

Success will be measured by the model's ability to correctly classify waste materials.

Week-by-Week Plan

 Week	Task	Milestone
Mar 16 - 22	Download dataset and setup environment	Dataset ready
Mar 23 - 29	Train initial classification model	Baseline model
Mar 30 - Apr 5	Improve accuracy with tuning	Improved Model
Apr 6 - 12	Build prediction demo	Demo Working
Apr 13 - 19	Testing and documentation	Project Complete
May 11 - 17	Final presentation	Presentation Ready

Challenges & Backup Plans

Challenge 1

Images may have different lighting conditions or backgrounds.

Plan B

Apply data augmentation and image normalization.

Challenge 1

Model accuracy may not be high enough.

Plan B:

Try a different pretrained model such as:

- ResNet18
- EfficientNet

Resources Needed

Resource	Tool
Compute	Google Colab
Framework	TensorFlow/Keras
Dataset	Kaggle Garbage Classification Dataset
Libraries	numpy, matplotlib, tensorflow
Estimated Cost	\$0

GitHub Repository

https://github.com/Iffraah96/ITAI1378_WasteSortingClassifier/tree/main